Toronto Bicycle Thefts Data Analysis

DAMG 7370

Team

Chaitanya Sai Gandi	Cloud Infra & ETL	EC2, Kafka, S3, Python ETL
Mudra Pandya	Data Modeling & SQL	RDS schema, SQL queries & views
Qi Han	BI & Dashboards	QuickSight visuals & storytelling

The Problem

Urban Challenge & Hidden Patterns

Bicycle theft is a persistent urban challenge in Toronto

Bikes are **easy to steal**, resell, and move across neighborhoods

Individual thefts reported, but **bigger picture** remains hidden

Raw police data (CSV) isn't actionable for decision-makers

Key Strategic Questions

Where are bikes stolen most often?

When does theft risk peak?

Which bike types/values are most targeted?

Why This Dataset?

Real-world Relevance

(genuine urban problem, helps actual stakeholders)

Rich Data Quality

(35K+ records, coordinates, multiple dimensions)

Perfect for Streaming

(event-level, time-stamped, open/free access)

Our Solution

Data Transformation & Source

Transform raw police data into interactive dashboards

- **Data:** Toronto Police Open Data (35,000+ theft records)

Empowering Target Users

Cyclists → understand personal risk

City planners → allocate prevention resources

Law enforcement → identify patterns & hotspots

AWS Services Used

EC2 (t3.xlarge)
Host Kafka, ZooKeeper, Python ETL

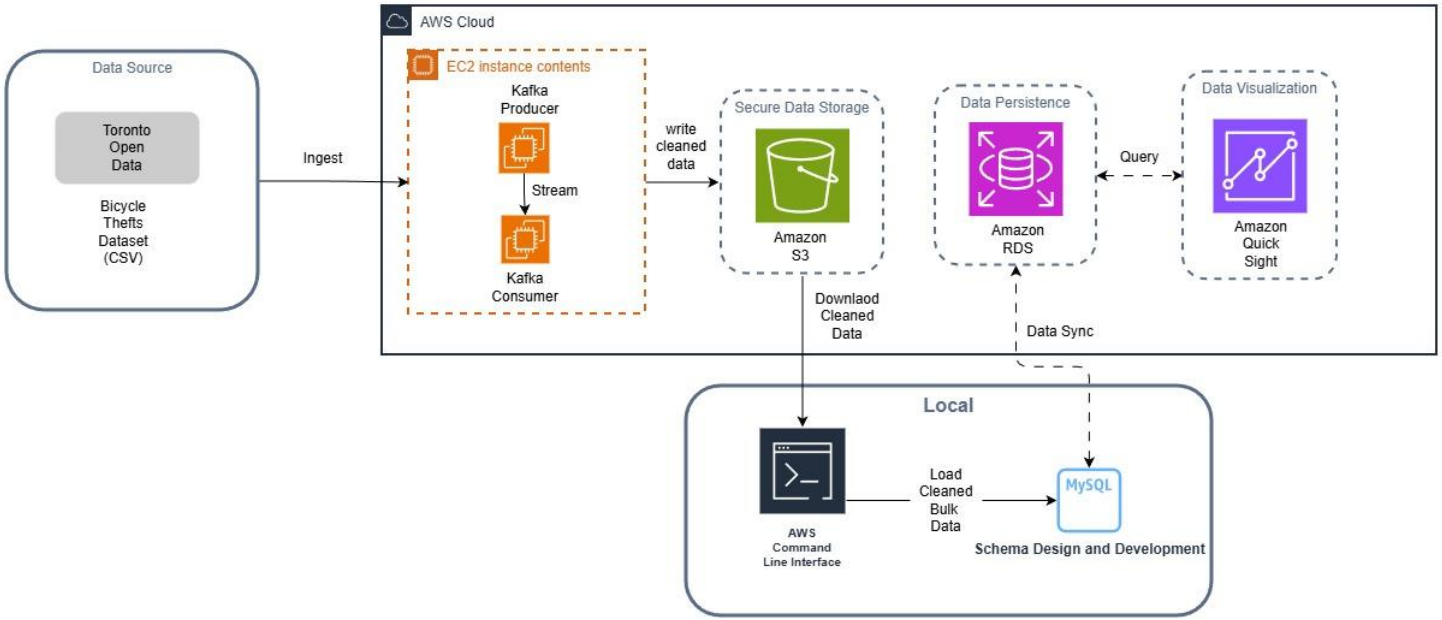
Apache Kafka
Message streaming layer

S3
Data lake for cleaned dataset

Amazon RDS
Relational Database Storage

QuickSight
Interactive dashboards

Architecture Overview



Why This Architecture?

EC2

Amazon EC2 (Compute & Control)

- Full root access for custom tuning (JVM, Kafka).
- Direct SSH debugging & config.
- Cost-effective single instance (t3.xlarge).

S3

Amazon S3 (Data Lake Storage)

- Cheap, durable object storage.
- Central repo for cleaned data.
- Easy collaboration & sharing.

RDS

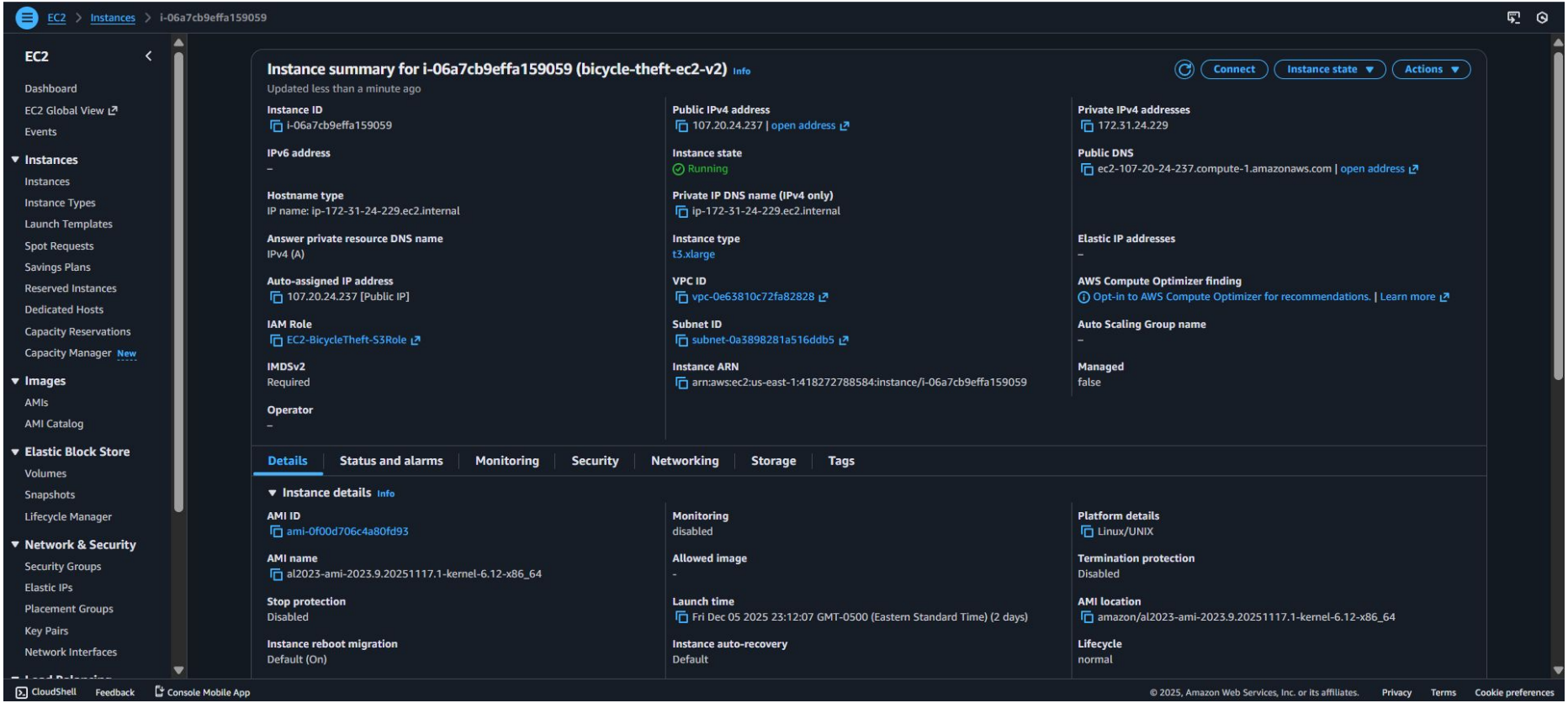
Amazon RDS MySQL (Structured Data Layer)

- Relational schema & SQL queries.
- Create views for analysis.
- Fine-grained access control.

QuickSight

Amazon QuickSight (Visual Analytics)

- Native AWS integration (no code).
- Simplified networking & auth.
- Focus on interactive visuals & storytelling.



This screenshot shows the EC2 instance (bicycle-theft-ec2-v2) that hosts our data processing stack. It is a t3.xlarge instance in us-east-1 with public IP 107.20.24.237 and IAM role EC2-BicycleTheft-S3Role. On this machine we installed Java, Kafka, ZooKeeper and Python, and we run the Kafka producer and consumer ETL scripts that read the raw bicycle theft CSV, stream it through Kafka, clean it with pandas, and upload the cleaned dataset to S3.

ETL Pipeline

Producer Script:

Consumer Script:

Data Cleaning Steps

- **Standardize columns** → lowercase, underscores
- **Parse dates** → handle invalid/missing values safely
- **Convert numerics** → coerce errors to NaN
- **Normalize text** → uppercase, strip whitespace
- Drop rows missing dates or coordinates
- Remove duplicate records
- Drop redundant geometry column

```
print(f"Reading raw CSV from: {INPUT_CSV}")
df = pd.read_csv(INPUT_CSV)

print("Raw DataFrame shape:", df.shape)
df.head()
```

[3] Python

```
Reading raw CSV from: D:\NEU\Fall 2025\DAMG 7370\Bicycle-Theft\bicycle-thefts-4326.csv
Raw DataFrame shape: (39473, 30)
```

	_id	EVENT_UNIQUE_ID	PRIMARY_OFFENCE	OCC_DATE	OCC_YEAR	OCC_MONTH	OCC_DOW	OCC_DAY	OCC_DOY	OCC_HOUR	...	BIKE_MAKE	BIKE_MODEL	BIKE_TYPE	BIKE_SPEED	BIKE_COLOUR	BIKE_COST	STATUS	LONG_WGS84	LAT_W
0	1	GO-20141263544	B&E	2013-12-26	2013	December	Thursday	26	360	19	...	FELT	F59	RC	21.0	SILRED	1300.0	STOLEN	-79.395643	43.6
1	2	GO-20141263784	PROPERTY - FOUND	2014-01-01	2014	January	Wednesday	1	1	18	...	TREK	SOHO S	RG	1.0	BLK	NaN	RECOVERED	-79.414654	43.6
2	3	GO-20141261431	THEFT UNDER	2014-01-01	2014	January	Wednesday	1	1	7	...	SUPERCYCLE	NaN	MT	10.0	NaN	NaN	STOLEN	-79.443645	43.6
3	4	GO-20149000090	THEFT UNDER	2014-01-01	2014	January	Wednesday	1	1	12	...	GI	TCX2 (2010)	OT	9.0	BLU	1019.0	STOLEN	-79.393760	43.6
4	5	GO-20141267465	THEFT UNDER	2013-09-30	2013	September	Monday	30	273	0	...	NORCO	CHARGER 9.2	MT	21.0	BLK	750.0	STOLEN	-79.404678	43.6

5 rows × 30 columns

Raw Toronto Bicycle Thefts dataset loaded in pandas

This cell shows the initial load of the `bicycle-thefts-4326.csv` file into a pandas DataFrame. The raw dataset has **39,473 rows and 30 columns**, including event details, dates, location context, bike attributes and geospatial fields. At this stage we can already see issues such as mixed data types and missing values in fields like `BIKE_TYPE` and `BIKE_COST`, which motivates the subsequent cleaning and standardization steps in our ETL pipeline.

```
[4]
df.columns
```

Python

```
Index(['_id', 'EVENT_UNIQUE_ID', 'PRIMARY_OFFENCE', 'OCC_DATE', 'OCC_YEAR',
       'OCC_MONTH', 'OCC_DOW', 'OCC_DAY', 'OCC_DOY', 'OCC_HOUR', 'REPORT_DATE',
       'REPORT_YEAR', 'REPORT_MONTH', 'REPORT_DOW', 'REPORT_DAY', 'REPORT_DOY',
       'REPORT_HOUR', 'DIVISION', 'LOCATION_TYPE', 'PREMISES_TYPE',
       'BIKE_MAKE', 'BIKE_MODEL', 'BIKE_TYPE', 'BIKE_SPEED', 'BIKE_COLOUR',
       'BIKE_COST', 'STATUS', 'LONG_WGS84', 'LAT_WGS84', 'geometry'],
      dtype='object')
```

```
[5]
# Standardize: lowercase, strip spaces, replace spaces with "_"
df.columns = [str(c).strip().lower().replace(" ", "_") for c in df.columns]

print("Renamed columns:")
df.columns
```

Python

```
Renamed columns:

Index(['_id', 'event_unique_id', 'primary_offence', 'occ_date', 'occ_year',
       'occ_month', 'occ_dow', 'occ_day', 'occ_doy', 'occ_hour', 'report_date',
       'report_year', 'report_month', 'report_dow', 'report_day', 'report_doy',
       'report_hour', 'division', 'location_type', 'premises_type',
       'bike_make', 'bike_model', 'bike_type', 'bike_speed', 'bike_colour',
       'bike_cost', 'status', 'long_wgs84', 'lat_wgs84', 'geometry'],
      dtype='object')
```

Standardizing column names for consistent downstream processing

In this step I inspect the original column names from the raw CSV and then standardize them by converting everything to lowercase, stripping extra spaces, and replacing spaces with underscores (e.g., `OCC_DATE` → `occ_date`, `REPORT_YEAR` → `report_year`). Having clean, consistent column names makes it much easier to write SQL, build views in MySQL, and reference fields later in the Kafka ETL code and QuickSight dashboards.

```python
def parse_date_safe(val):
    """Parse date strings safely; return NaT on failure."""
    if pd.isna(val) or val == "":
        return pd.NaT
    try:
        return parser.parse(str(val), dayfirst=False, yearfirst=False)
    except Exception:
        return pd.NaT
```

[6] Python

```python
date_cols = ["occ_date", "report_date"]

for col in date_cols:
    if col in df.columns:
        df[col] = df[col].apply(parse_date_safe)

df[date_cols].dtypes
```

[7] Python

```
occ_date       datetime64[ns]
report_date    datetime64[ns]
dtype: object
```

```python
# Quick look at parsed dates
df[date_cols].head()
```

[8] Python

	occ_date	report_date
0	2013-12-26	2014-01-01
1	2014-01-01	2014-01-01
2	2014-01-01	2014-01-01
3	2014-01-01	2014-01-02
4	2013-09-30	2014-01-02

Converting raw date strings into proper datetime fields

Here I define a helper function `parse_date_safe` that safely converts raw date strings into pandas `datetime` objects and returns `NaT` for any invalid or missing values. I then apply this function to the two key date columns, `occ_date` and `report_date`. The output shows that both columns are now stored as `datetime64` and the preview confirms that the dates are parsed correctly, which is important for doing time-series analysis (by year, month, day, or hour) later in SQL and QuickSight.

```python
numeric_cols = [
    "_id",
    "occ_year",
    "occ_day",
    "occ_doy",
    "occ_hour",
    "report_year",
    "report_day",
    "report_doy",
    "report_hour",
    "bike_speed",
    "bike_cost",
    "long_wgs84",
    "lat_wgs84",
]

for col in numeric_cols:
    if col in df.columns:
        df[col] = pd.to_numeric(df[col], errors="coerce")

df[numeric_cols].dtypes
```

[9] Python

```
_id             int64
occ_year        int64
occ_day         int64
occ_doy         int64
occ_hour        int64
report_year     int64
report_day      int64
report_doy      int64
report_hour     int64
bike_speed    float64
bike_cost     float64
long_wgs84    float64
lat_wgs84     float64
dtype: object
```

Converting key fields to numeric types

In this step I define a list of columns that should contain numeric values (years, day-of-week indexes, hours, bike speed/cost, and latitude/longitude). I then loop through this list and use `pd.to_numeric(..., errors="coerce")` to force each column into a numeric type, converting any invalid values to `NaN`. The dtype summary at the bottom confirms that these fields are now stored as `int64` or `float64`, which is required for reliable aggregations, statistics, and mapping in MySQL and QuickSight.

```python
# Check a few numeric columns
df[numeric_cols].head()
```

[10] Python

	_id	occ_year	occ_day	occ_doy	occ_hour	report_year	report_day	report_doy	report_hour	bike_speed	bike_cost	long_wgs84	lat_wgs84
0	1	2013	26	360	19	2014	1	1	17	21.0	1300.0	-79.395643	43.640021
1	2	2014	1	1	18	2014	1	1	18	1.0	NaN	-79.414654	43.660525
2	3	2014	1	1	7	2014	1	1	7	10.0	NaN	-79.443645	43.637657
3	4	2014	1	1	12	2014	2	2	20	9.0	1019.0	-79.393760	43.642772
4	5	2013	30	273	0	2014	2	2	12	21.0	750.0	-79.404678	43.648964

```python
text_cols = [
    "event_unique_id",
    "primary_offence",
    "occ_month",
    "occ_dow",
    "report_month",
    "report_dow",
    "division",
    "location_type",
    "premises_type",
    "bike_make",
    "bike_model",
    "bike_type",
    "bike_colour",
    "status",
]

for col in text_cols:
    if col in df.columns:
        df[col] = (
            df[col]
            .astype(str)
            .str.strip()
            .str.upper()
        )

df[text_cols].head()
```

[11] Python

	event_unique_id	primary_offence	occ_month	occ_dow	report_month	report_dow	division	location_type	premises_type	bike_make	bike_model	bike_type	bike_colour	status
0	GO-20141263544	B&E	DECEMBER	THURSDAY	JANUARY	WEDNESDAY	D14	OTHER COMMERCIAL / CORPORATE PLACES (FOR PROFI...	COMMERCIAL	FELT	F59	RC	SILRED	STOLEN
1	GO-20141263784	PROPERTY - FOUND	JANUARY	WEDNESDAY	JANUARY	WEDNESDAY	D14	SINGLE HOME, HOUSE (ATTACH GARAGE, COTTAGE, MO...	HOUSE	TREK	SOHO S	RG	BLK	RECOVERED
2	GO-20141261431	THEFT UNDER	JANUARY	WEDNESDAY	JANUARY	WEDNESDAY	D14	APARTMENT (ROOMING HOUSE, CONDO)	APARTMENT	SUPERCYCLE	NAN	MT	NAN	STOLEN
3	GO-20149000090	THEFT UNDER	JANUARY	WEDNESDAY	JANUARY	THURSDAY	D52	APARTMENT (ROOMING HOUSE, CONDO)	APARTMENT	GI	TCX2 (2010)	OT	BLU	STOLEN

Standardizing text columns for consistent analysis

Here I define a list of all categorical/text columns (offence details, dates as words, division, location and premises type, bike make/model/type/colour, and status). For each of these columns I convert the values to string, strip extra spaces, and convert everything to uppercase. The preview at the bottom shows the cleaned text (e.g., months and days in uppercase and without trailing spaces). This makes grouping, filtering, and joining in MySQL and QuickSight much more reliable and avoids issues caused by inconsistent spelling or casing.

```python
# Before dropping, check missing values
df[["occ_date", "lat_wgs84", "long_wgs84"]].isna().sum()
```

[12] Python

```
occ_date       0
lat_wgs84    341
long_wgs84   341
dtype: int64
```

```python
# Drop rows without occurrence date
if "occ_date" in df.columns:
    df = df.dropna(subset=["occ_date"])

# Drop rows missing coordinates
for coord_col in ["lat_wgs84", "long_wgs84"]:
    if coord_col in df.columns:
        df = df.dropna(subset=[coord_col])

print("Shape after dropping missing occ_date / coordinates:", df.shape)
```

[13] Python

```
Shape after dropping missing occ_date / coordinates: (39132, 30)
```

```python
if "geometry" in df.columns:
    df = df.drop(columns=["geometry"])

before = df.shape[0]
df = df.drop_duplicates()
after = df.shape[0]

print(f"Rows before drop_duplicates: {before}")
print(f"Rows after drop_duplicates:  {after}")
print("Current shape:", df.shape)
```

[14] Python

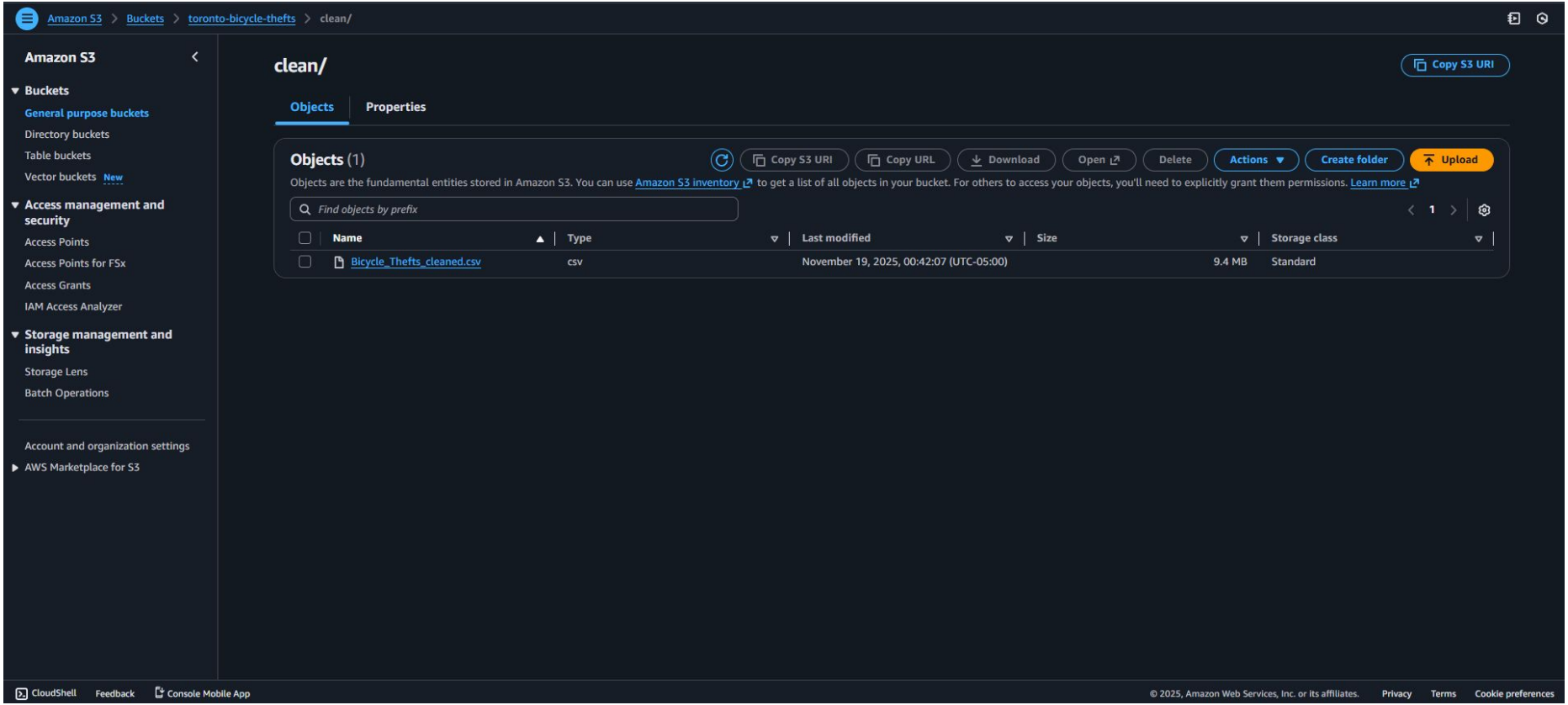
```
Rows before drop_duplicates: 39132
Rows after drop_duplicates:  39132
Current shape: (39132, 29)
```

Removing rows with missing key fields and duplicates

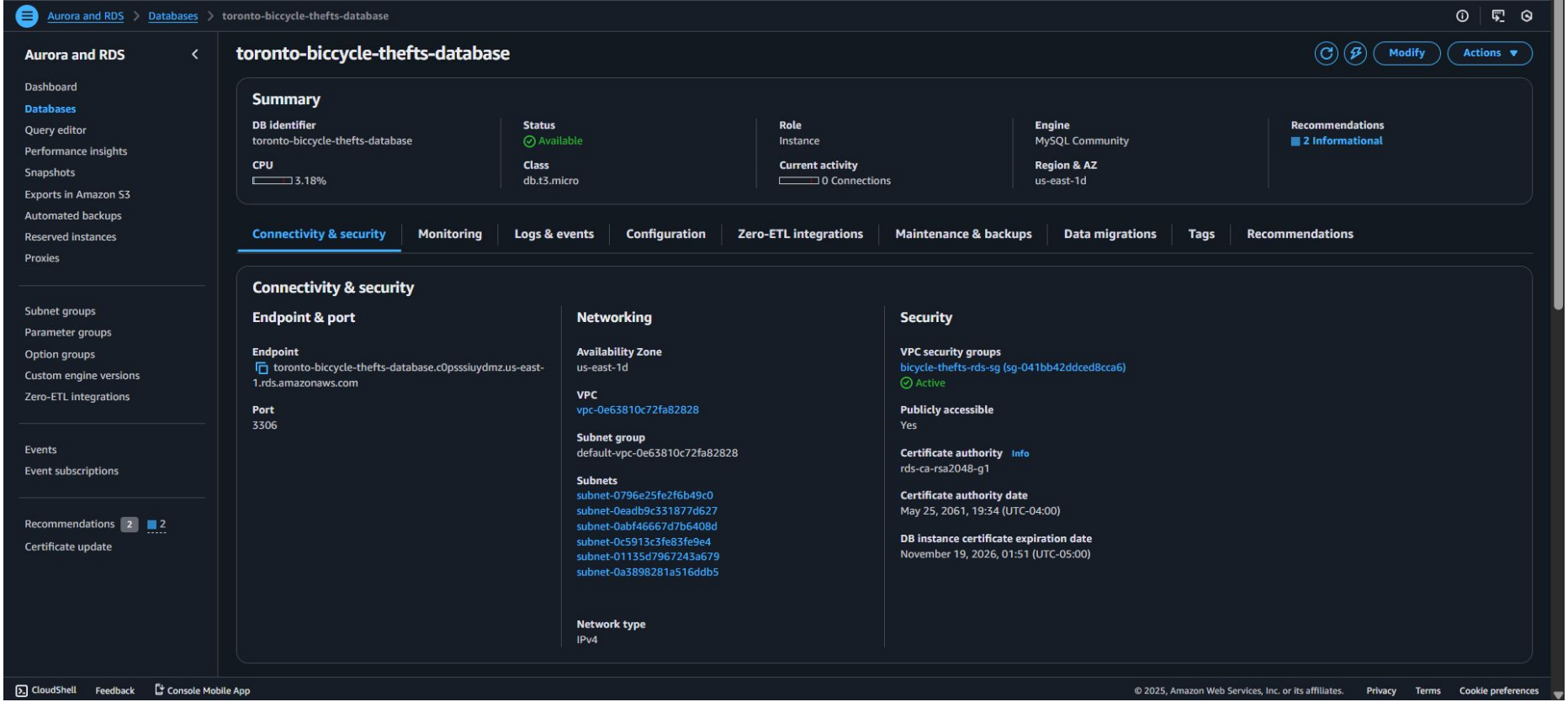
In this step I first check how many missing values we have in the critical columns `occ_date`, `lat_wgs84`, and `long_wgs84`. Then I drop any rows that do not have an occurrence date or valid latitude/longitude, because those records cannot be used for time-based or map-based analysis. After that I remove the `geometry` column (it stores the same location in a complex JSON format, which we don't need once we have clean lat/long) and call `drop_duplicates()` to ensure that duplicate incidents are not counted twice. The printout at the bottom shows how many rows remain and confirms the final cleaned shape of the dataset.

Challenges & Solutions

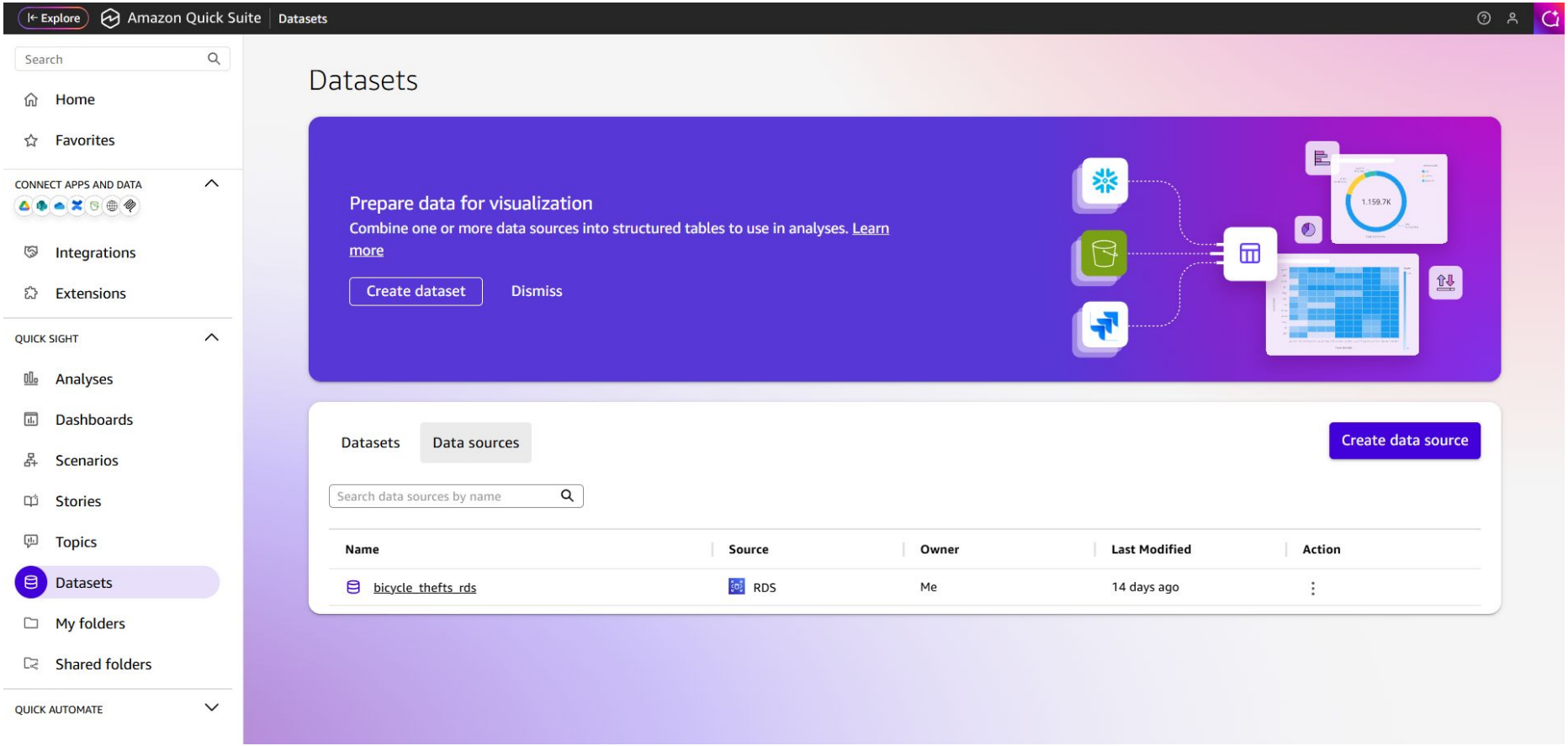
THE CHALLENGES	THE SOLUTIONS
EC2 memory limits	Upgraded to t3.xlarge
Kafka connectivity	Configured advertised.listeners
Cross-account S3 access	Added bucket policy
Data quality issues	Robust pandas cleaning



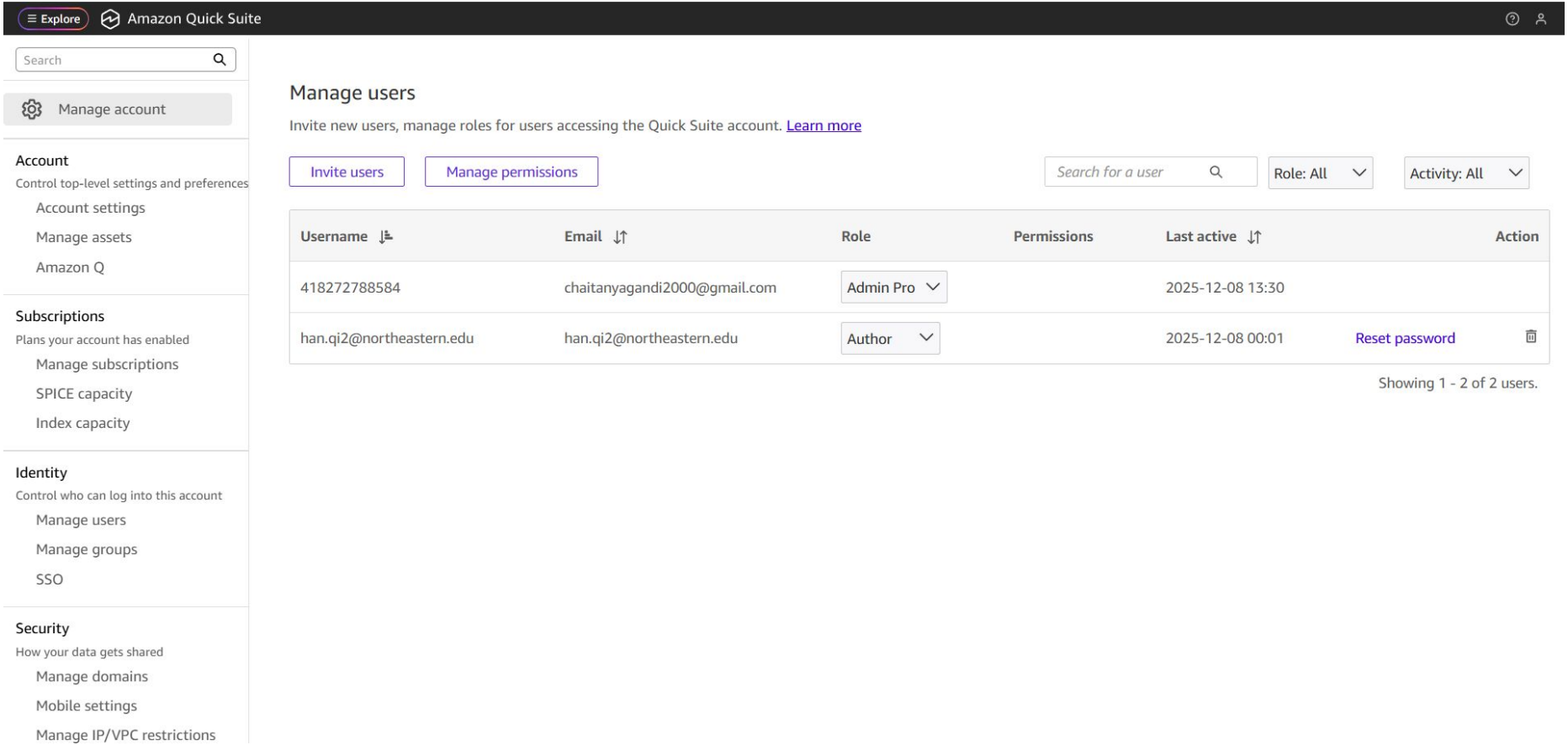
This screenshot shows the `toronto-bicycle-thefts` S3 bucket, specifically the `clean/` prefix that contains the processed file `Bicycle_Thefts_cleaned.csv`. This object is the output of the Kafka consumer ETL job running on EC2. S3 acts as our data lake: the cleaned CSV in this location is shared with teammates (via bucket policy) so they can download it, load it into Amazon RDS MySQL, and use it as the source for SQL analysis and QuickSight dashboards.



This screenshot shows the `toronto-bicycle-thefts-database` Amazon RDS instance that we use as the relational analytics layer. The engine is **MySQL Community** (db.t3.micro) running in `us-east-1d`, exposed on port **3306** with public access enabled so teammates can connect from MySQL Workbench and QuickSight. The endpoint shown here is used to load the cleaned CSV from S3 and create tables/views, and later used as the data source for building the Toronto bicycle theft dashboards.



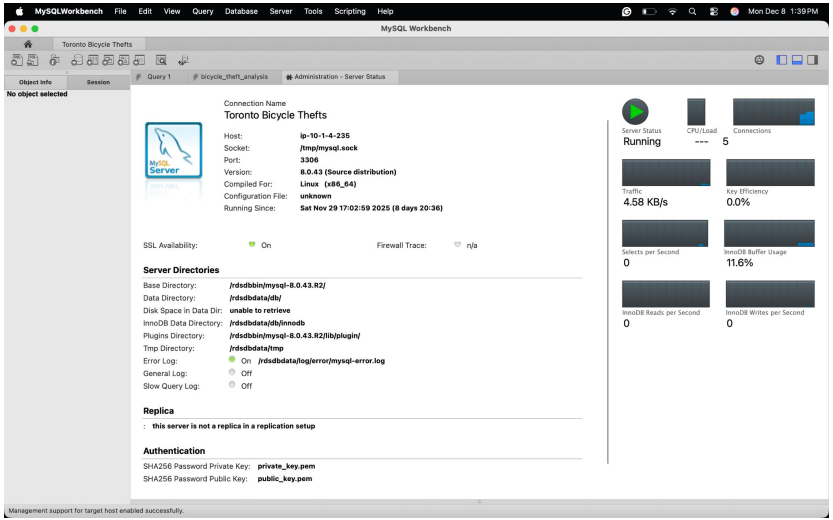
In this step, I created an Amazon QuickSight account to build the dashboards for our project and added a data source that points directly to our RDS MySQL instance. Because QuickSight is natively integrated with RDS, connecting the database and starting to design charts, maps, and filters was straightforward and required almost no extra setup.

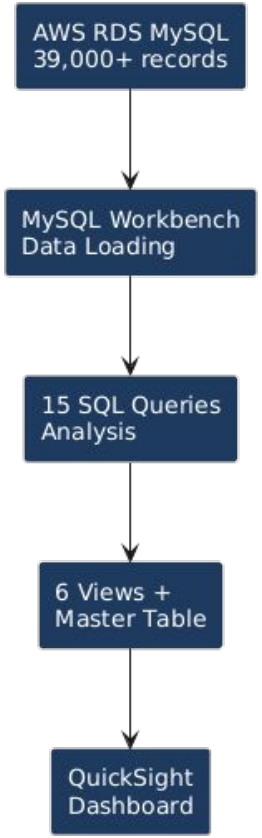


In QuickSight, I also configured access for my team. From **Manage account** → **Manage users**, I invited my teammate using their email and assigned them the **Author** role. This ensures they can log in, connect to the shared RDS data source, and independently build and edit the project dashboards while I keep admin control of the account.

SQL Analysis & Database Design

- Connected to AWS RDS MySQL database
- Loaded 39,000+ bicycle theft records
- Developed 15 analytical queries
- Created 6 views + master table for dashboards



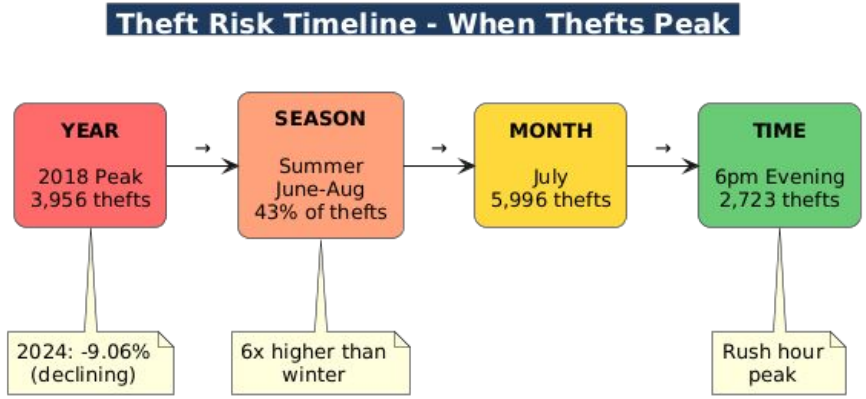


When Do Thefts Happen? - Temporal Analysis

SQL Query: Analyzed theft trends over time

Key Findings:

- **Peak Year:** 2018 with 3,956 thefts
- **2024 Trend:** Declining! Down 9.06% from 2023
- **Peak Month:** July (5,996 thefts - summer cycling season)
- **Peak Hour:** 6pm (2,723 thefts - evening commute)
- **Highest Risk Period:** Evening (6pm-10pm)



Result Grid Filter Rows: Search

occ_year	total_thefts	percentage
2014	3031	7.75
2015	3288	8.40
2016	3810	9.74
2017	3872	9.89
2018	3956	10.11
2019	3720	9.51
2020	3886	9.93
2021	3153	8.06
2022	2949	7.54
2023	2981	7.62
2024	2711	6.93

hour_of_day	theft_count	time_period
18	2723	Evening (6pm-10pm)
17	2540	Afternoon (12pm-6p…
12	2365	Afternoon (12pm-6p…
19	2282	Evening (6pm-10pm)
0	2204	Night (10pm-6am)
16	2112	Afternoon (12pm-6p…
20	2107	Evening (6pm-10pm)
9	2054	Morning (6am-12pm)
15	2007	Afternoon (12pm-6p…
21	1892	Evening (6pm-10pm)

Where Do Thefts Happen? - Geographic Analysis

SQL Query: Analyzed theft locations and recovery rates

Key Findings:

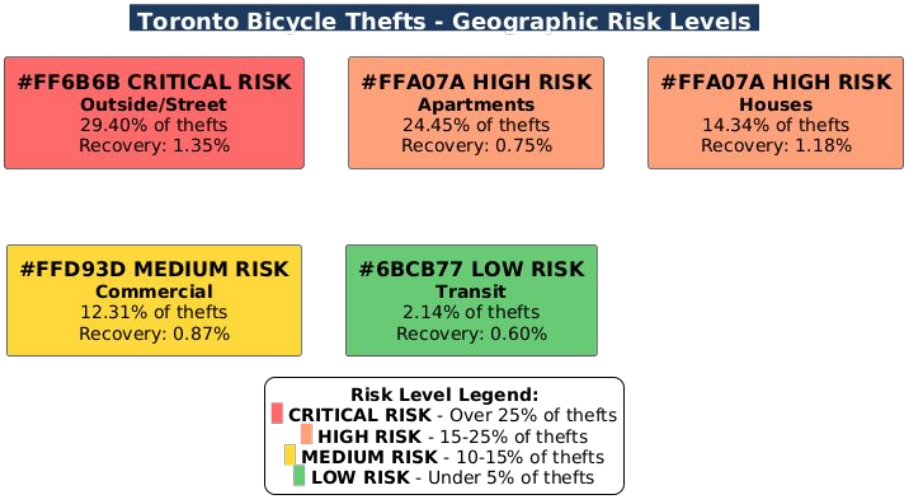
- **Highest Risk:** Outside/Street parking (29.40% of all thefts)
- **Second Highest:** Apartments (24.45%)
- **Recovery Crisis:** ALL locations show <2% recovery rate
- **Best Police Division:** D31 with only 1.64% recovery
- **Worst Division:** D32 with 0.59% recovery
- **Parking Lots:** 9.34% of thefts - also high risk

```
72      location_type,
73      COUNT(*) as theft_count,
74      ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM bicycle_thefts WHERE location_type IS NOT NULL), 2) as percentage,
75      ROUND(AVG(bike_cost), 2) as avg_bike_value
76  FROM bicycle_thefts
```

100% 47:75

Result Grid Filter Rows: Search Export:

location_type	theft_count	percentage	avg_bike_value
APARTMENT (ROOMING HOUSE, CONDO)	9567	24.45	1102.61
BANK AND OTHER FINANCIAL INSTITUTION...	258	0.66	1058.56
BAR / RESTAURANT	880	2.25	1153.68
COMMERCIAL DWELLING UNIT (HOTEL, MO...	119	0.30	1025.10
COMMUNITY GROUP HOME	8	0.02	519.71
CONSTRUCTION SITE (WAREHOUSE, TRAIL...	44	0.11	1281.84
CONVENIENCE STORES	372	0.95	998.15
DEALERSHIP (CAR, MOTORCYCLE, MARINE,...	23	0.06	1611.10
GAS STATION (SELF, FULL, ATTACHED CON...	34	0.09	824.04
GO BUS	9	0.02	774.29
GO STATION	310	0.79	652.74
GO TRAIN	16	0.04	621.80
GROUP HOMES (NON-PROFIT, HALFWAY HO...	14	0.04	1067.40
GROUP HOMES (NON-PROFIT, HALFWAY HO...	36	0.09	876.12
HOMELESS SHELTER / MISSION	57	0.15	1485.55
HOSPITAL / INSTITUTIONS / MEDICAL FACILI...	411	1.05	1024.71
JAILS / DETENTION CENTRES	2	0.01	400.00
NURSING HOME	108	0.28	993.94
OPEN AREAS (LAKES, PARKS, RIVERS)	725	1.85	897.36
OTHER COMMERCIAL / CORPORATE PLACE...	3086	7.89	1201.78
OTHER NON COMMERCIAL / CORPORATE P...	501	1.28	1031.63
OTHER PASSENGER TRAIN	1	0.00	300.00
OTHER PASSENGER TRAIN STATION	8	0.02	706.25
OTHER REGIONAL TRANSIT SYSTEM VEHIC...	3	0.01	775.00
OTHER TRAIN ADMIN OR SUPPORT FACILITY	2	0.01	390.00
OTHER TRAIN TRACKS	1	0.00	[illegible]
OTHER TRAIN YARD	2	0.01	1300.00
PARKING LOTS (APT., COMMERCIAL OR NO...	3654	9.34	1141.96
PHARMACY	695	1.78	988.81
POLICE / COURTS (PAROLE BOARD, PROBA...	58	0.15	965.27
PRIVATE PROPERTY STRUCTURE (POOL, S...	2932	7.49	1184.30
RELIGIOUS FACILITIES (SYNAGOGUE, CHU...	57	0.15	1180.40



What Gets Stolen? - Bike Brand & Seasonal Analysis

SQL Query: Analyzed most stolen bike brands and seasonal patterns

Key Findings:

Bike Brands:

- **Most stolen:** OT brand (7,879 thefts, avg value $1,021)
- **Second:** UK brand (4,209 thefts, avg value $887)
- **Premium brands:** Specialized ($1,880 avg) shows 2.97% recovery
- **945 bikes** with "UNKNOWN" make - highlights registration need

Seasonal Patterns:

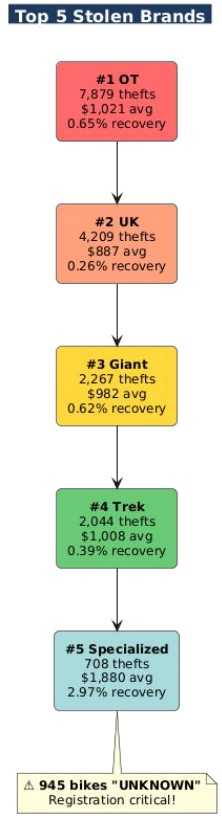
- **Summer (June-Aug):** 16,894 thefts (43% of annual)
- **Winter (Dec-Feb):** Only 3,229 thefts
- **6x higher risk** in summer vs winter
- **Winter bikes:** Higher average value ($1,300-$1,400)

```sql
-- 8. Most stolen bike makes (brands thieves target)
SELECT
    bike_make,
    COUNT(*) as times_stolen,
    ROUND(AVG(bike_cost), 2) as avg_value,
    COUNT(CASE WHEN status = 'RECOVERED' THEN 1 END) as recovered,
    ROUND(COUNT(CASE WHEN status = 'RECOVERED' THEN 1 END) * 100.0 / COUNT(*), 2) as recovery_rate
FROM bicycle_thefts
WHERE bike_make IS NOT NULL
  AND bike_make != ''
  AND bike_make != 'UNKNOWN'
```

100% 8:97

Result Grid | Filter Rows: Search | Export: | Fetch rows:

bike_make	times_stolen	avg_value	recovered	recovery_rate
OT	7879	1021.76	51	0.65
UK	4209	887.57	11	0.26
GI	2267	982.58	14	0.62
OTHER	2172	1492.28	34	1.57
TR	2044	1008.26	8	0.39
NO	1275	822.87	6	0.47
UNKNOWN MAKE	945	966.70	12	1.27
CC	859	419.10	1	0.12
TREK	844	1545.46	16	1.90
SU	832	270.73	0	0.00
GIANT	804	1261.60	19	2.36
CA	782	1397.46	4	0.51
RA	728	579.93	6	0.82
SPECIALIZED	708	1880.00	21	2.97
SC	705	498.78	0	0.00



Data Products for Dashboards

SQL Views & Master Table

SQL Views Created for QuickSight:

I built **6 analytical views** to enable interactive dashboards:

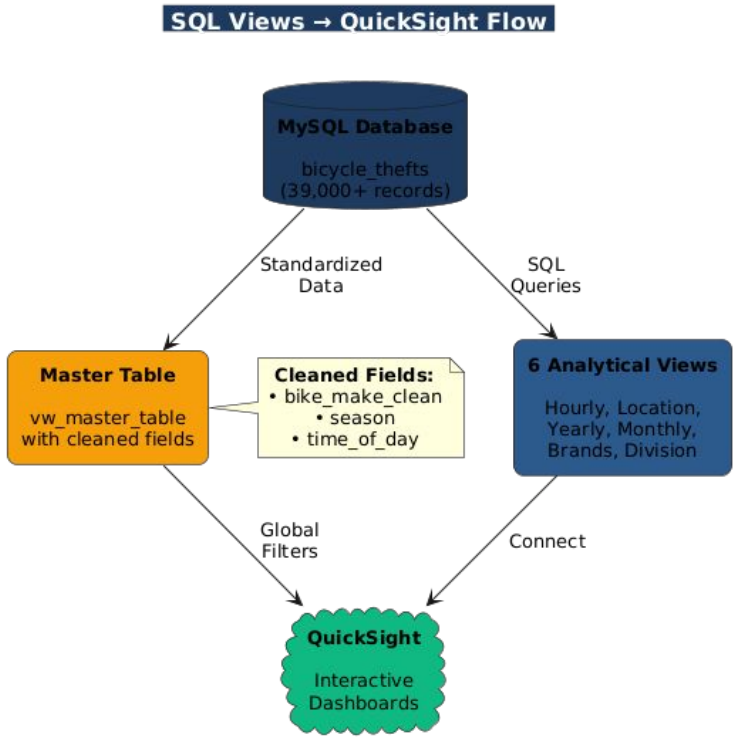
- **vw_hourly_thefts** - Time pattern analysis
- **vw_top_locations** - Geographic hotspots with recovery rates
- **vw_yearly_trends** - Year-over-year theft trends
- **vw_monthly_patterns** - Seasonal analysis with bike values
- **vw_bike_brands** - Brand targeting with recovery metrics
- **vw_division_performance** - Police division performance

Master Table (vw_master_table):

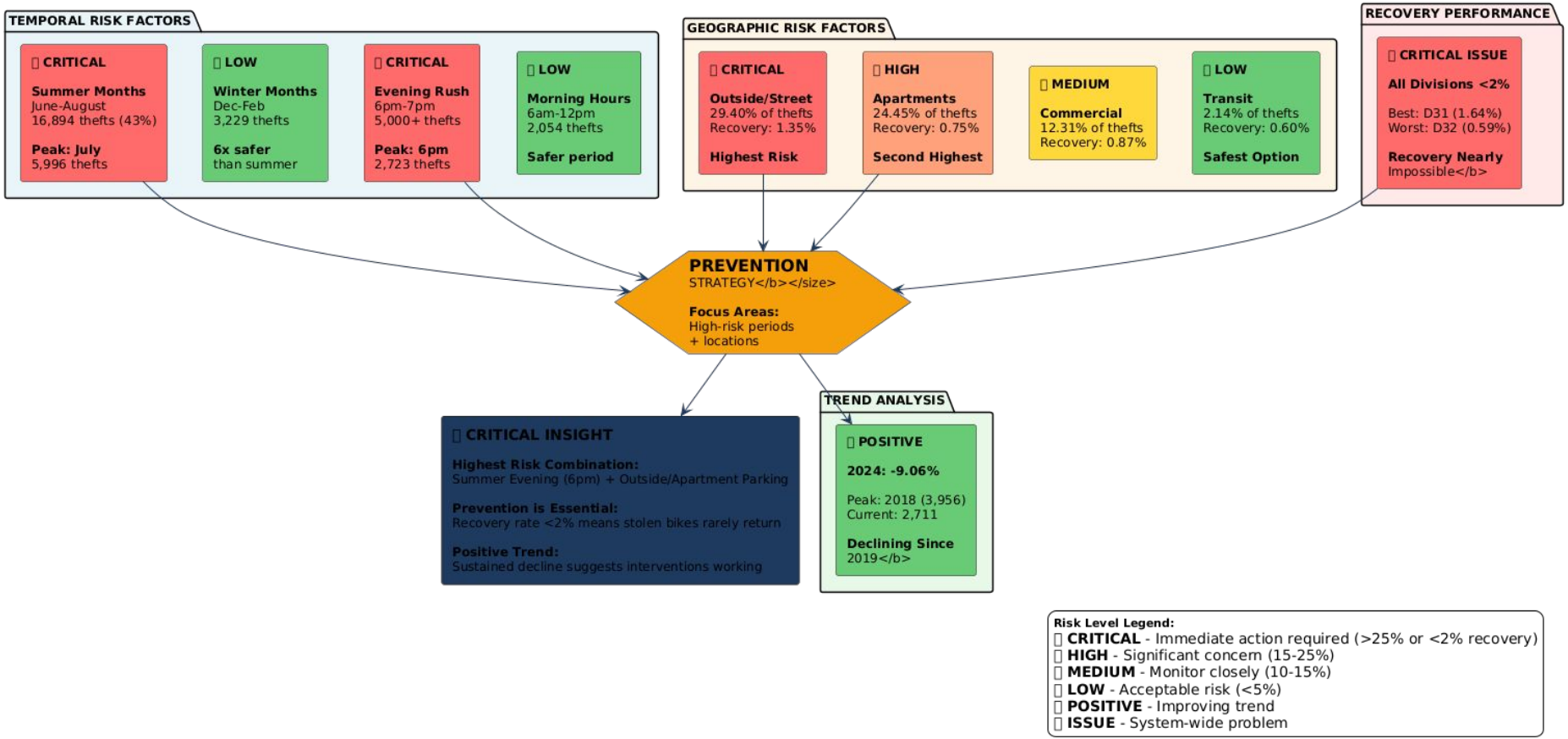
Built for interactive filtering and global dashboards:

- **bike_make_clean** - Standardized brand names (GIANT, TREK, etc.)
- **season** - Winter/Spring/Summer/Fall classification
- **time_of_day** - Morning/Afternoon/Evening/Night periods

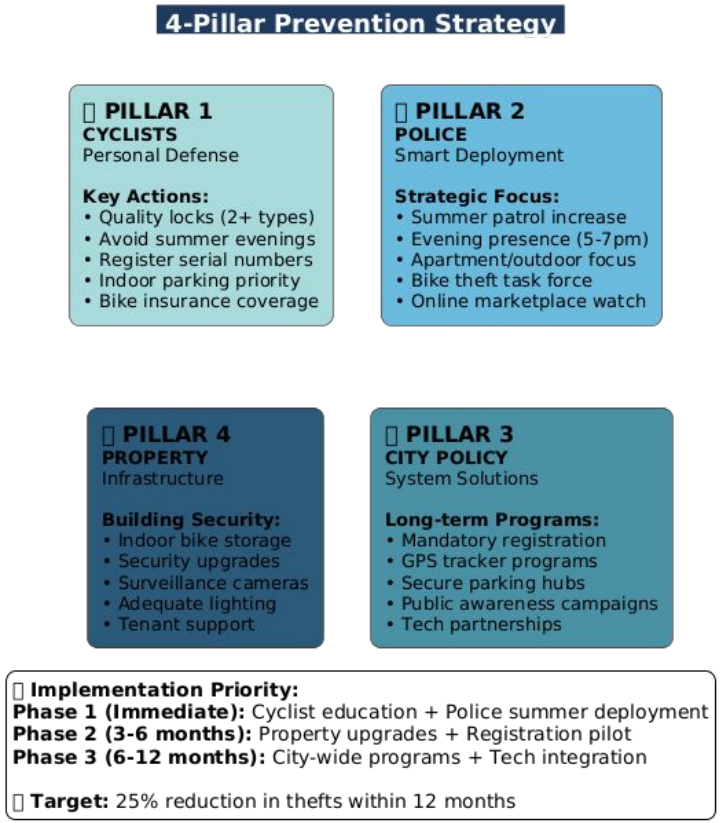
Purpose: Enable Qi to create dynamic dashboards with filters



Toronto Bicycle Thefts - Comprehensive Risk Assessment Matrix



Recommendations & Action Plan



Why This Strategy Works:

Data-Driven Approach:

- Analysis identified clear patterns in when, where, and what gets stolen
- Each pillar targets a specific high-risk factor from our findings

Multi-Stakeholder Solution:

- No single group can solve alone - requires coordinated effort
- Addresses all risk dimensions: temporal, geographic, recovery

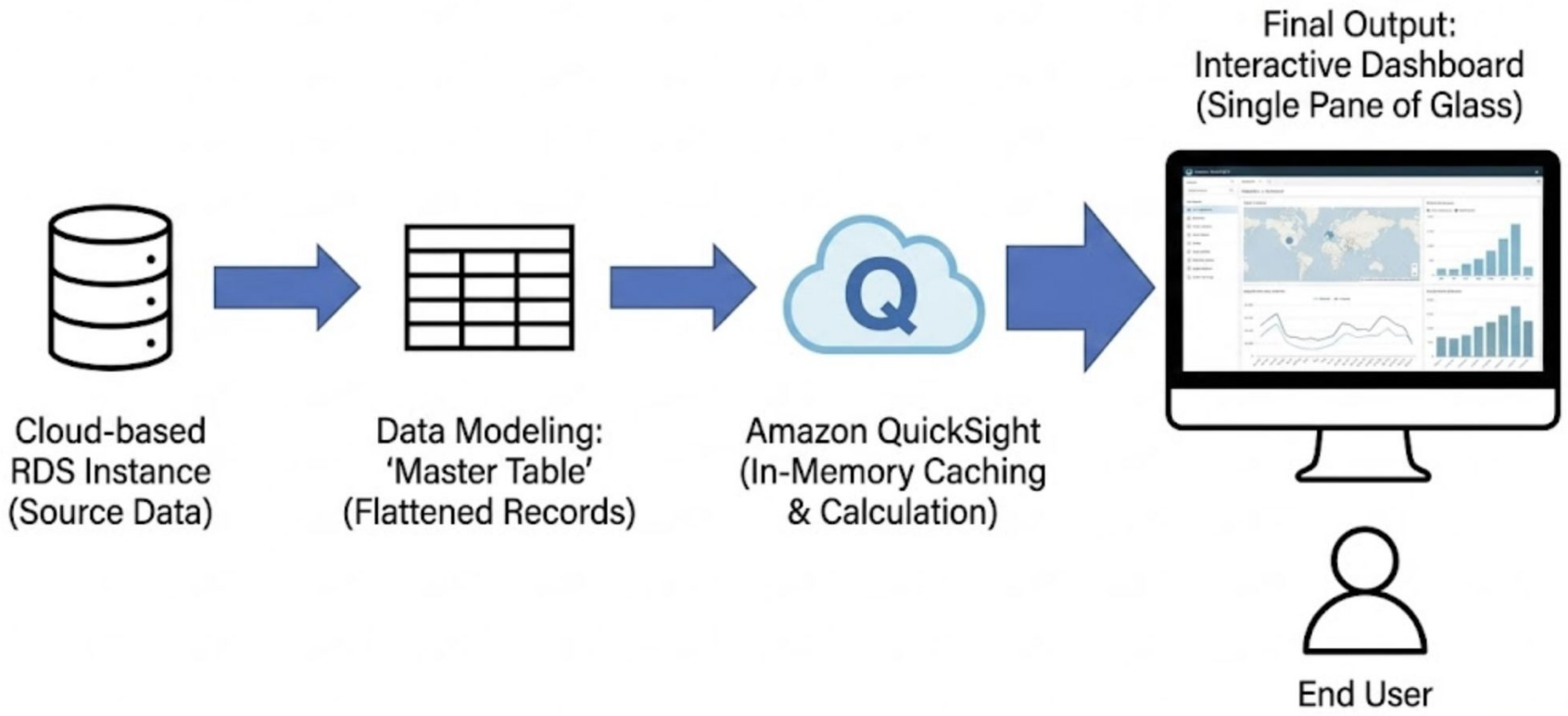
Proven Success Potential:

- 2024 shows 9.06% decline - interventions are working
- Strategic focus can accelerate this positive trend

Prevention Over Recovery:

- Current recovery rate: <2% across all divisions
- Prevention is the only viable strategy
- Coordinated approach maximizes impact

BI Implementation & Visual Analytics



Technical Architecture & Backend Strategy

BI Platform Selection: Amazon QuickSight (Cloud-native integration with RDS).

Data Modeling Strategy:

- Moved from fragmented SQL views to a single **"Master Table" architecture**.
- Enables dynamic aggregation and flexible filtering on the fly.

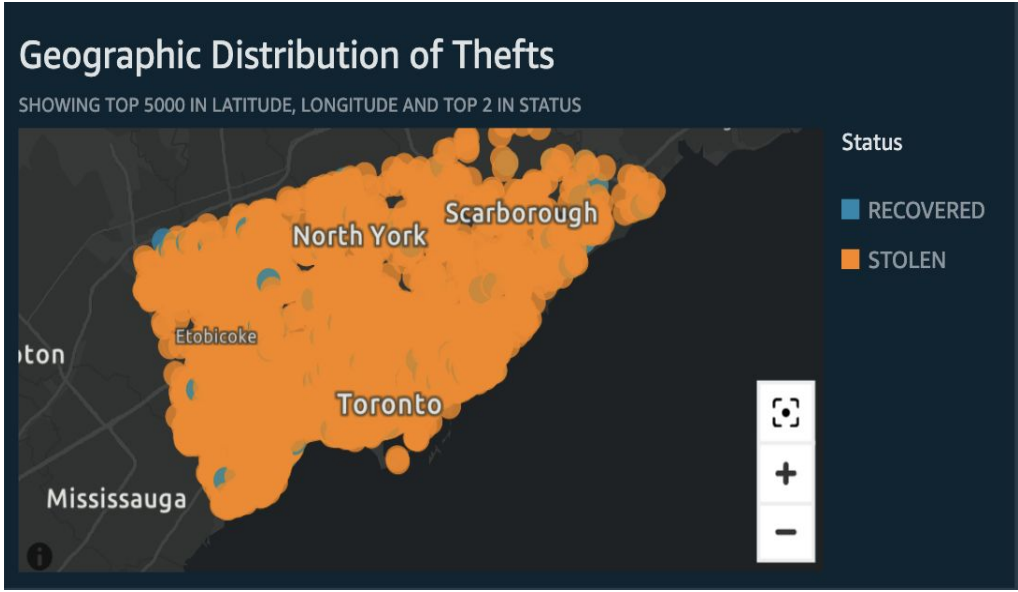
Performance Optimization (SPICE):

- Utilized **S**uper-fast, **P**arallel, **I**n-memory **C**alculation **E**ngine.
- **Result:** Sub-second load times for 10+ years of geospatial data.

Visual Analysis – The "Where" and "When"

Spatial Analysis (The Map):

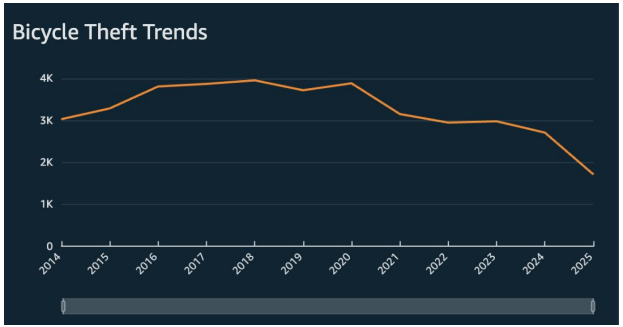
- **Visualization Type:** Interactive **Point Map** (Pin-level Granularity).
- **Implementation:** Plotted individual theft coordinates (Latitude/Longitude) to visualize exact incident locations rather than aggregated regions.
- **Insight Logic:** Applied a conditional color schema (**Orange = Stolen**, **Blue = Recovered**), allowing users to identify precise "recovery hotspots" amidst the high-density theft zones.

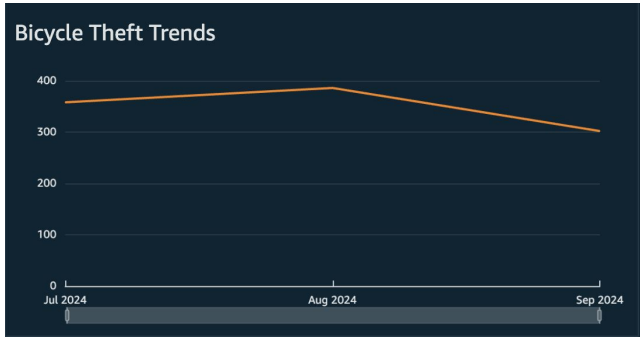


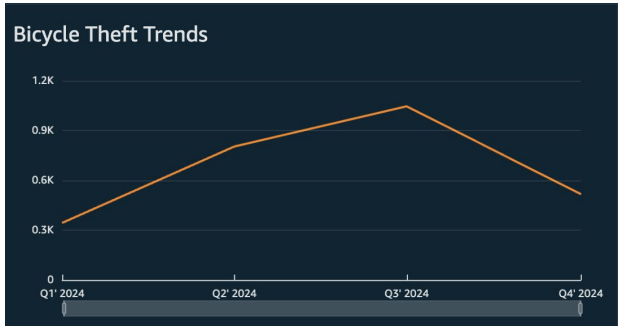
Visual Analysis – The "Where" and "When"

Temporal Analysis (The Trend):

- **Analysis Type:** 10-Year Time Series tracking theft volume.
- **Interactive Feature:** Implemented a **Drill-Down Hierarchy (Year/Quarter/Month)**.
- **Why it matters:** Allows users to switch between viewing long-term annual trends and investigating specific seasonal "waves" (Summer spikes) without changing the chart view.





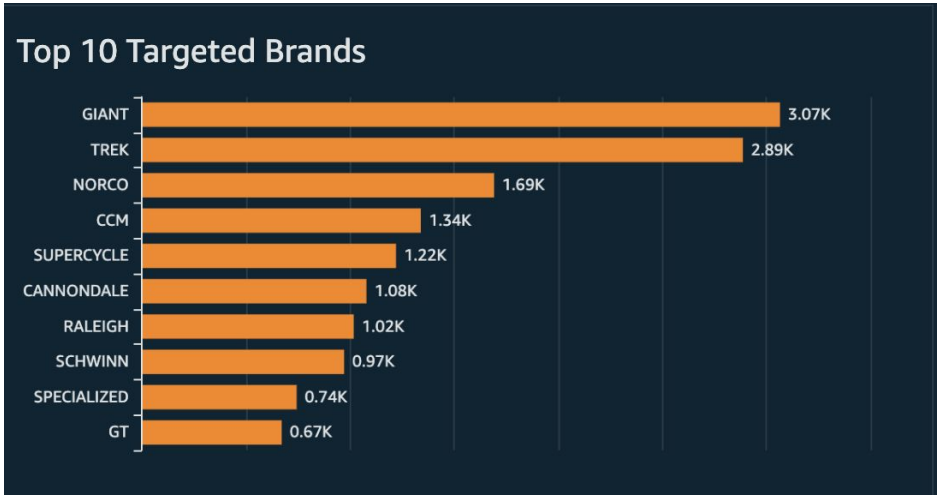


Behavioral Risk & Target Profiling

- **Behavioral Risk (The Heatmap):**
  - **Method:** Matrix crossing *Day of Week* vs. *Time of Day*.
  - **Discovery:** Isolates "Danger Zones" (e.g., Friday Evenings) that simple line charts miss.

Behavioral Risk & Target Profiling

- **Target Analysis (The Bar Chart):**
  - **Method:** Top 10 Ranking Filter on `Bike_Make_Clean`.
  - **Constraint:** Filtered "Unknown" data to focus on actionable insights.
  - **Insight:** Identifies premium brands (Giant, Trek) as primary targets for organized theft.



Dashboard Design & User Experience

"Single Pane of Glass" Architecture:

- Implemented **Free-form layout** with auto-scaling ("Fit to Window"), ensuring the entire dashboard is visible without scrolling on any device.

Global Filter Suite:

- Five interconnected controls (**Year, Neighborhood, Bike Type, Day, Time**) allow users to slice the entire dataset instantly.

Omni-Directional Interactivity:

- **Unified Action Layer:** Interactivity is enabled on **all components**, including the **4 KPI Cards**, Map, Trend Line, Heatmap, and Bar Chart.
- **Context-Aware Metrics:** Clicking any visual (e.g., a map cluster) filters all other charts and **instantly recalculates KPI totals** to reflect the selected data.

Conclusion & Critical Insights

1. The Good News: Declining Trend

- Bicycle thefts dropped over **9%** in 2024, continuing a multi-year downward trajectory.

2. The Bad News: Recovery Crisis

- The recovery rate remains extremely low at **less than 2%** across all police divisions.

3. High-Risk Locations

- **Apartments** are the most vulnerable premises type, accounting for nearly **25%** of all reported thefts.

4. "Danger Zones" (Time & Season)

- Risk is not random. **Summer months** see **6x more thefts** than winter, peaking specifically on July evenings.

5. Targeted Assets

- Thieves are brand-aware. Premium brands like **Giant and Trek** are primary targets, while "Unknown" bikes still account for significant losses.

THANK YOU